



2026/27 SEMESTER 1 – TEST

Course : Academic Support & Enrichment Programme

Unit : ITS105 – Complementary Applied Maths in Computing

Date: *20 July 2026*

Name : _____ Admin No : _____

Unit Group : ITS105-?

Score

100

INSTRUCTION:

- Answer all questions.
- Write your answers in the provided space or type them in a softcopy document.
- If handwritten, scan or convert your answers to PDF/image format.
- Submit all answers to the Brightspace *ITS105 test submission* folder.

MARKING SCHEME

Solution for Differentiation (30 marks)

1. Given a function $f(x) = (2x^2 + 1)^3$. Find its derivative when $x = 1$. (10 marks)

Answer:

$$f'(x) = 3(2x^2 + 1)^2 (4x)$$

$$\text{When } x=1, f'(1) = 3(3)^2 (4) = 108$$

2. Given a function $f(x) = (x^3 - x^2 + 1)(x^2 + 2)$. Find its derivative when $x = 1$. (10 marks)

Answer:

$$(3x^2 - 2x)(x^2 + 2) + (x^3 - x^2 + 1)(2x) = 5x^4 - 4x^3 + 6x^2 - 2x$$

$$\text{When } x = 1, f'(1) = 5 - 4 + 6 - 2 = 5$$

3. Given a function $f(x) = \frac{x^2-1}{x-2}$. Find its derivative when $x = 0$. (10 marks)

Answer:

$$f'(x) = \frac{(x-2)(2x) - (x^2-1)(1)}{(x-2)^2} = \frac{x^2-4x+1}{(x-2)^2}$$

$$f'(0) = \frac{1}{4}$$

Solution for Integration (30 marks)

4. Find the integral $\int x^6 dx$

(10 marks)

Answer:

$$\frac{x^7}{7} + C$$

5. Find the integral $\int \frac{1}{\sqrt[3]{x^2}} dx$

(10 marks)

Answer:

$$3\sqrt[3]{x} + C$$

6. Evaluate the definite integral $\int_{-1}^2 (6x + 9x^2 - 8) dx$

(10 marks)

Answer:

$$= \left[\frac{6x^2}{2} + \frac{9x^3}{3} - 8x \right]_{-1}^2$$

$$= [3x^2 + 3x^3 - 8x]_{-1}^2$$

$$= [3(2)^2 + 3(2)^3 - 8(2)] - [3(-1)^2 + 3(-1)^3 - 8(-1)]$$

$$= 12$$

Solution for Applications (40 marks)

7. Find the equation of tangent line to the curve $y = x^2 - 4$ at $x = 4$. (10 marks)

Answer:

$y = x^2 - 4$, $dy/dx = 2x$
 when $x = 4$, gradient m of tangent line to the curve $= dy/dx = 2 \times 4 = 8$
 when $x = 4$, $y = 4^2 - 4 = 12$. Hence coordinates $(4, 12)$.

Equation of tangent line $(y - 12) / (x - 4) = 8$
 $\Rightarrow y = 8x - 20$

8. Given $y = x^3 - 12x$, find the coordinates of the local minima and local maxima. (10 marks)

Answer:

- i. $dy/dx = 3x^2 - 12$
 ii. Find critical point when $dy/dx = 0$

$$3x^2 - 12 = 0$$

$$x = +2 \text{ or } -2$$

- iii. Find y to get the coordinates

When $x = 2$, $y = -16$
 When $x = -2$, $y = 16$

Critical points are: $(2, -16)$, $(-2, 16)$

- iv. Investigate the intervals $x < -2$, $-2 < x < 2$, $x > 2$

When $x < -2$, $dy/dx = +ve$
 When $-2 < x < 2$, $dy/dx = -ve$
 When $x > 2$, $dy/dx = +ve$

- v. Determine local maxima and local minima

$(-2, 16)$ is local maxima
 $(2, -16)$ is local minima

9. Find the area bounded by the curve $y = -x^2 + 4x$ and the x-axis ($y = 0$).

(10 marks)

Answer:

Intersection: $-x^2 + 4x = 0 \Rightarrow x = 0 \text{ or } x = 4$

$$\begin{aligned} & \int_0^4 -x^2 + 4x \, dx \\ &= \left[\frac{-x^3}{3} + 2x^2 \right]_0^4 \\ &= -\frac{64}{3} + 32 \\ &= \frac{32}{3} \quad (\text{or } 10.67) \end{aligned}$$

10. A cyclist moves along a straight line. Its velocity $v(t)$ at time t (in seconds) is given by the function: $v(t) = 3t^2 - 4t + 5$ where $v(t)$ is in meters per second. Calculate the total distance traveled by the particle during the first 5 seconds of its motion.

Note: the distance travelled D is given by the integral: $D = \int_{t_1}^{t_2} v(t) \, dt$.

(10 marks)

Answer:

$$\begin{aligned} D &= \int_0^5 (3t^2 - 4t + 5) \, dt \\ &= [t^3 - 2t^2 + 5t + C]_0^5 \\ &= (125 - 50 + 25) - 0 \\ &= 100 \end{aligned}$$

The total distance travelled by the particle during the first 5 seconds of its motion is 100 meters.

End of Paper